

INSERTION SORT

$A = [23, 1, 10, 5, 2, 4]$

temp
5

$A = [1, 23, 10, 5, 2, 4]$

temp = A[i]

$A = [1, 10, 23, 5, 2, 4]$

j = i - 1

$A = [1, 5, 10, 23, 2, 4]$

while (j >= 0 and A[j] > temp)

{
A[j+1] = A[j]

j--;
}

$A = [1, 2, 5, 10, 23, 4]$

A[j+1] = temp;

1
↓
2
↓
5
↓
10
↓
23
↓
4

$A = [$
 $\begin{array}{cccccc}
& & 8 & 10 & 10 & 10 \\
& 1 & 10 & 10 & 10 & 10 \\
2 & 2 & 2 & 2 & 2 & 2 \\
0 & 1 & 2 & 3 & 4 & 5
\end{array}$
 $]$



temp
4

for ($i = 1$; $i < N$; $i++$)

{

temp = $A[i]$;

$j = i - 1$;

while ($j \geq 0$ and $A[j] > temp$)

{

$A[j+1] = A[j]$;

$j--$;

}

$A[j+1] = temp$;

}

MERGE Two SORTED ARRAYS

A = [1, 7, 8, 150, 160] N

B = [2, 5, 6, 9, 18, 19, 59] M

int C[N+M];

C = [1, 7, 8, 150, 160, 2, 5, 6, 9, 18, 19, 59]

↓

[1, 2, 5, 6, 7, 8, 9, 18, 19, 59, 150, 160]

$$A = [\underset{0}{1}, \underset{1}{7}, \underset{2}{8}, \underset{3}{150}, \underset{4}{160}] \quad \overset{5}{}$$

$$B = [\underset{0}{2}, \underset{1}{5}, \underset{2}{6}, \underset{3}{9}, \underset{4}{18}, \underset{5}{19}, \underset{6}{519}] \quad \overset{7}{}$$

$$C = [\underset{0}{\underline{1}}, \underset{1}{\underline{2}}, \underset{2}{\underline{5}}, \underset{3}{\underline{6}}, \underset{4}{\underline{7}}, \underset{5}{\underline{8}}, \underset{6}{\underline{9}}, \underset{7}{\underline{18}}, \underset{8}{\underline{19}}, \underset{9}{\underline{150}}, \underset{10}{\underline{160}}, \underset{11}{\underline{519}}] \quad \overset{k}{}$$

int A[N];

// input

int B[M];

// input

```
int C[N+M];
```

```
int i = 0, j = 0, k = 0;
```

```
while (i < N and j < M)
```

```
{
```

```
    if (A[i] < B[j])
```

```
    {
```

```
        C[k] = A[i];
```

```
        i++;
```

```
    } else
```

```
    {
```

```
        C[k] = B[j];
```

```
        j++;
```

```
    }
```

```
    k++;
```

```
}
```

$A = [10, 20]$ ⁱ

$B = [7, 8, 15]$ ^j

$C = [\underline{7} \ \underline{8} \ \underline{10} \ \underline{15} \ \underline{20}]$ ^k

while ($i < N$)

{

$C[k] = A[i];$

$i++;$

$k++;$

}

while ($j < M$)

{

$C[k] = B[j];$

$j++;$

$k++;$

}

